Individual Contribution Report

Student Name: John Benedict Tuvilla

Group: MiniOS Omega (1063)

Assigned Role: Process and Memory Developer

What did I contribute to the project?

I coded the Process Management and Memory Management parts of MiniOS. I made sure the OS could correctly keep track of running programs and manage the system's memory.

What features did I personally develop?

I created the commands for processes and memory. For processes, I made the commands to list them (process), start them (run), and stop them (kill). For memory, I made the commands to check the memory status (memory), give memory to a task (allocate), and take it back (free).

What problems did I encounter?

When connecting my code to the main shell, I got errors because the user inputs weren't being passed correctly. Also, keeping track of the total memory was hard. At first, my code was resetting the total memory instead of subtracting from it, so the system showed the wrong amount of free space.

How did I solve those problems?

I worked with my group to fix how the main shell sends inputs, making sure all my functions received the data the same way. To fix the memory bug, I made sure all my commands updated one main variable for the total memory, instead of creating new local variables that messed up the math.

What did I learn from this project?

I learned how a real operating system manages a computer's parts. Making the Process IDs (PIDs) taught me how an OS keeps track of multiple programs at once. Working on the memory commands taught me how memory is given out and taken back. It also helped me get much better at Python and linking different files together.

Evidence of my contribution:

You can see my work in the processes.py and memory.py files in our group's GitHub repository. You can also see where my commands are connected to the main shell inside the main.py file.